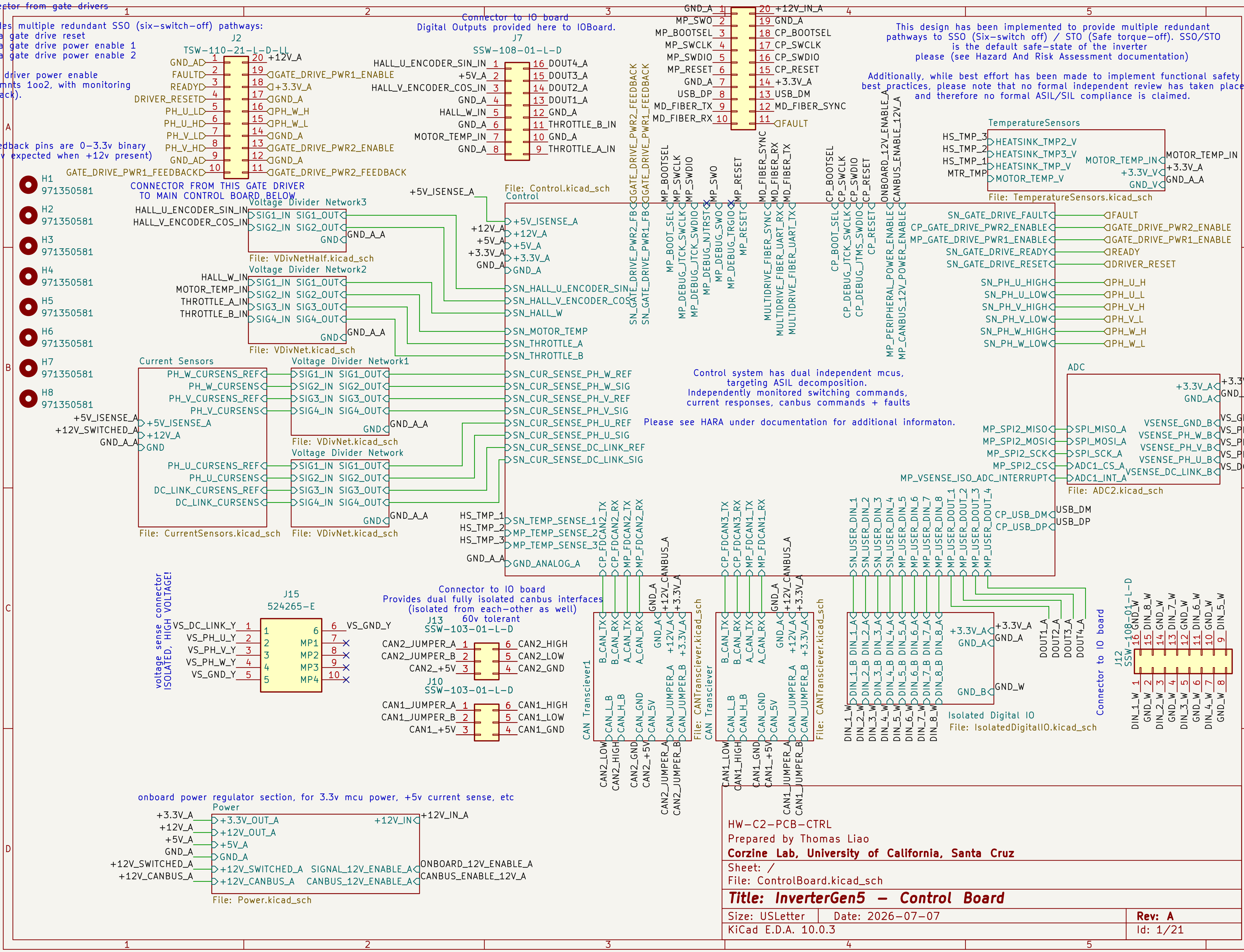
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD

NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION, DISABLE TIM1 SYNC PIN (TIM1 ETR)



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /

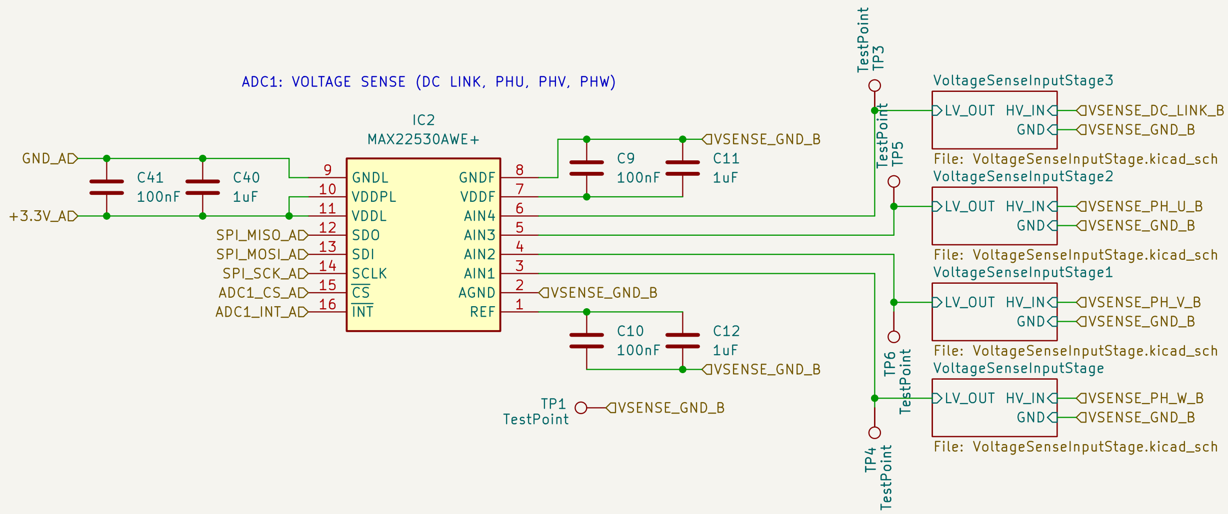
File: ControlBoard.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

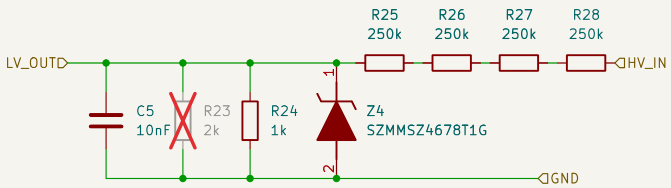
KiCad E.D.A. 10.0.3 | Id: 1/21

ADC1: VOLTAGE SENSE (DC LINK, PHU, PHV, PHW)



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/
File: ADC2.kicad_sch
Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 4/21

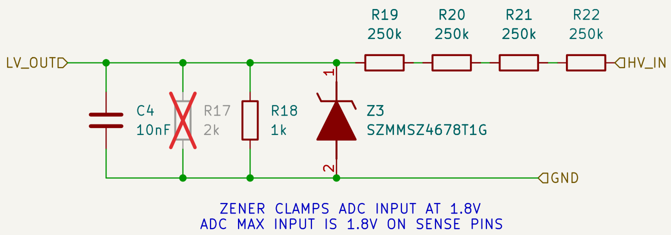


ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

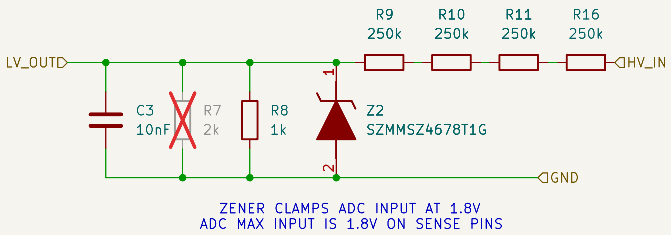
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch
Title: InverterGen5 — Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 3/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

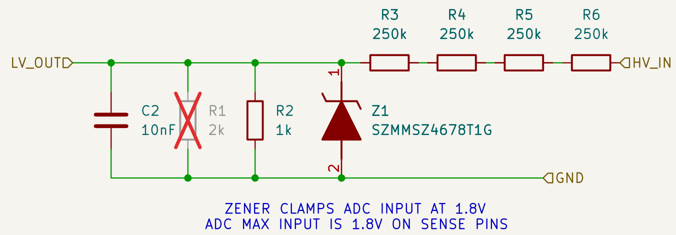
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21



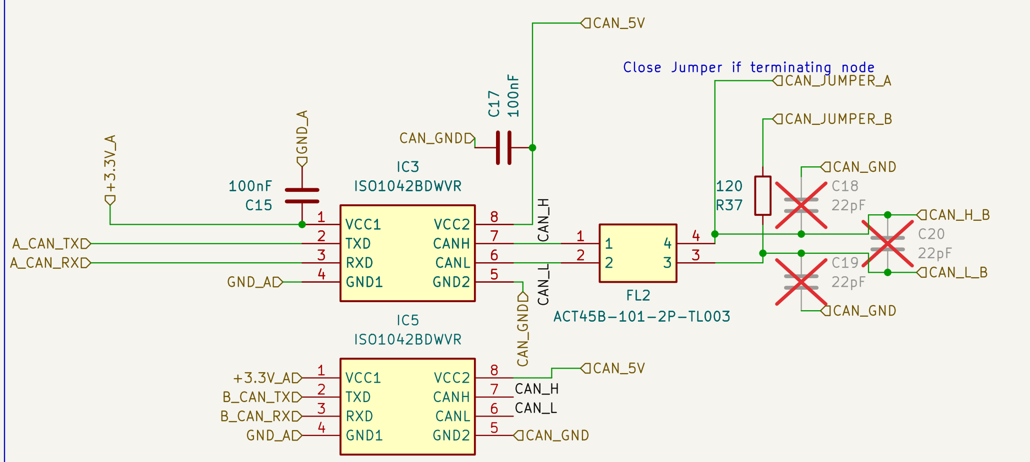
HW-C2-PCB-CTRL

Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

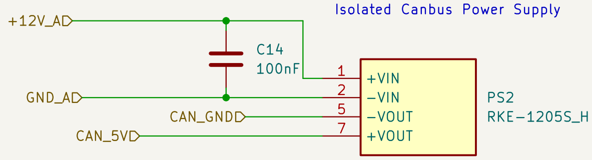
Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 6/21



two can transcievers, allows safety coproc to snoop and interact with can traffic, if needed.
provides redundant path to read and cross-check motor torque commands, if using canbus for torque cmd over throttle A/B inpts

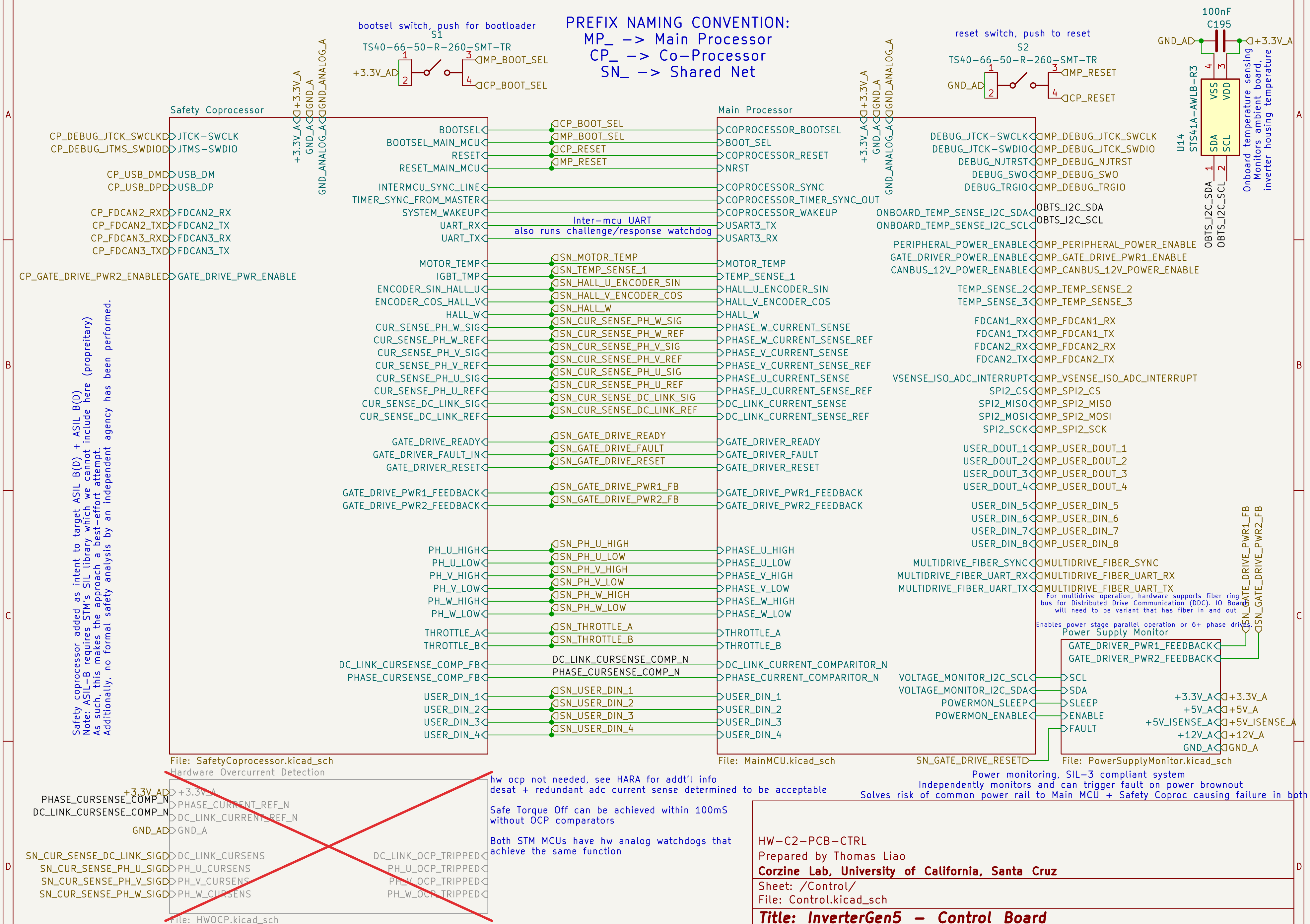
Isolated Canbus Power Supply



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 7/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

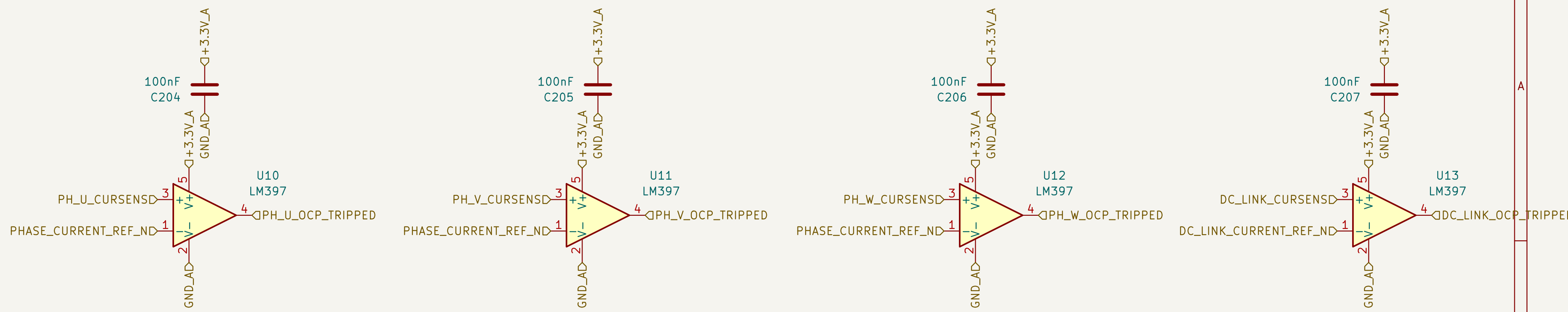
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/

File: Control.kicad_sch

Title: InverterGen5 — Control Board

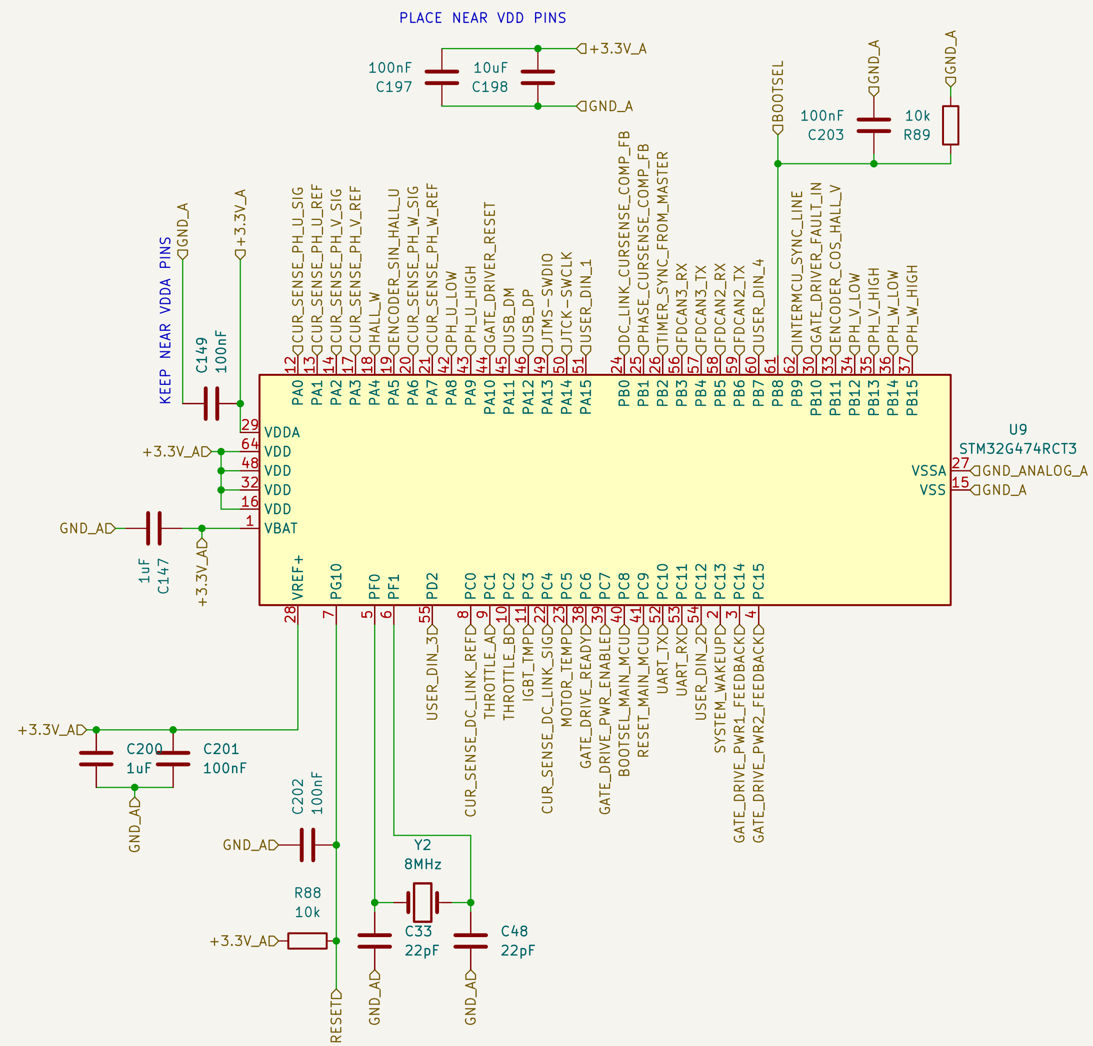
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 8/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch

Title: InverterGen5 – Control Board

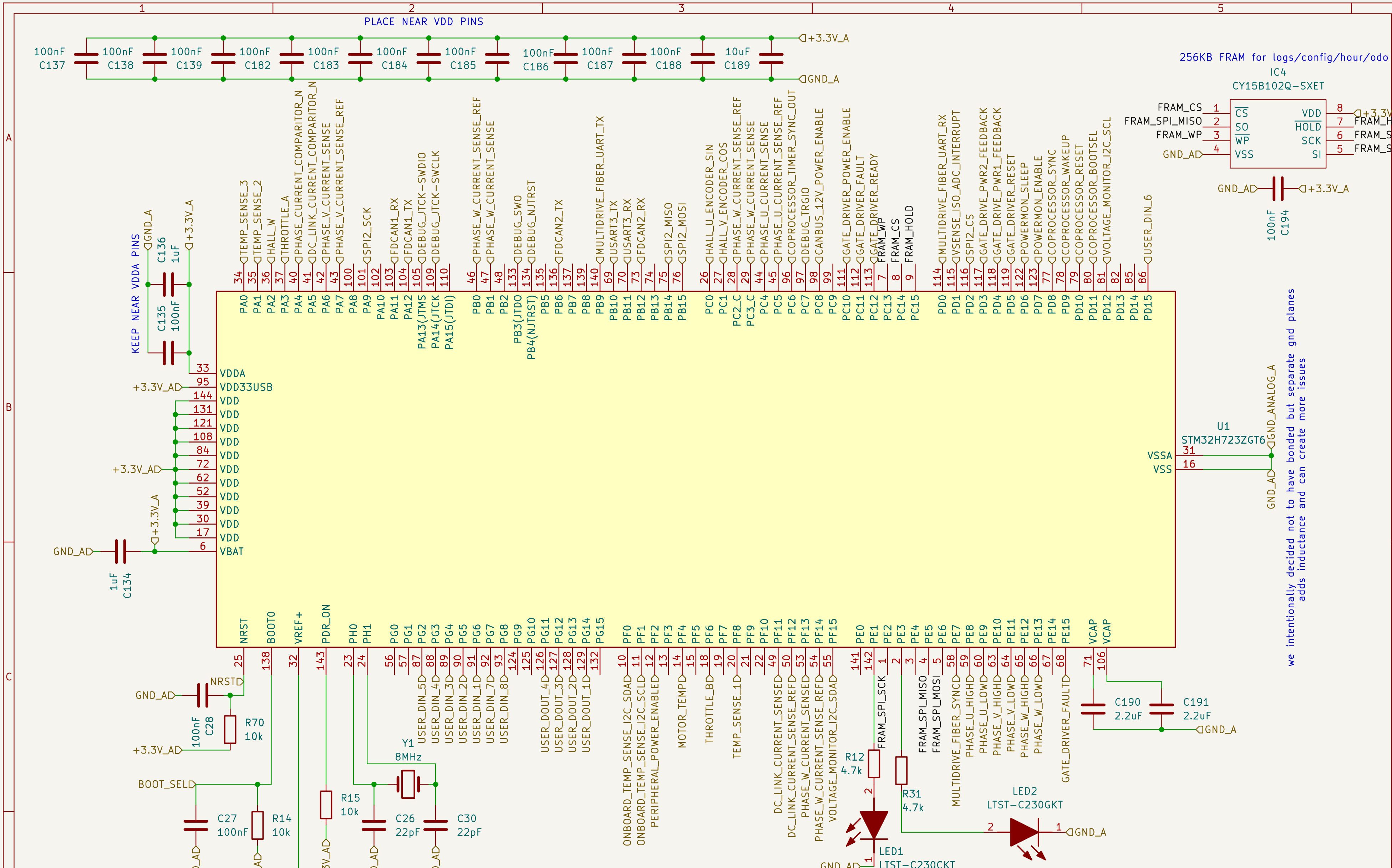
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 10/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Safety Coprocessor/
File: SafetyCoprocessor.kicad_sch

Title: InverterGen5 – Control Board

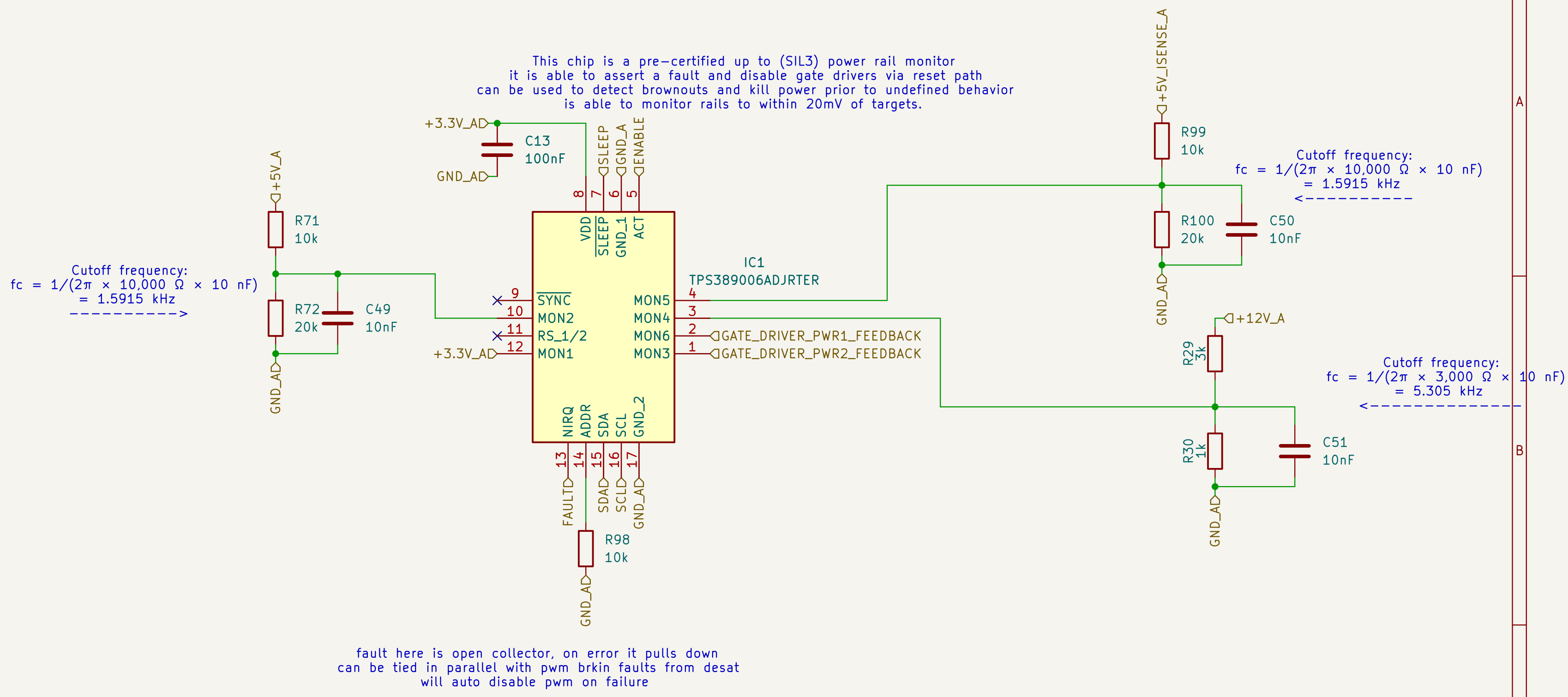
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 16/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Main Processor/
File: MainMCU.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 19/21



HW–C2–PCB–CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Power Supply Monitor/
File: PowerSupplyMonitor.kicad_sch

Title: InverterGen5 – Control Board

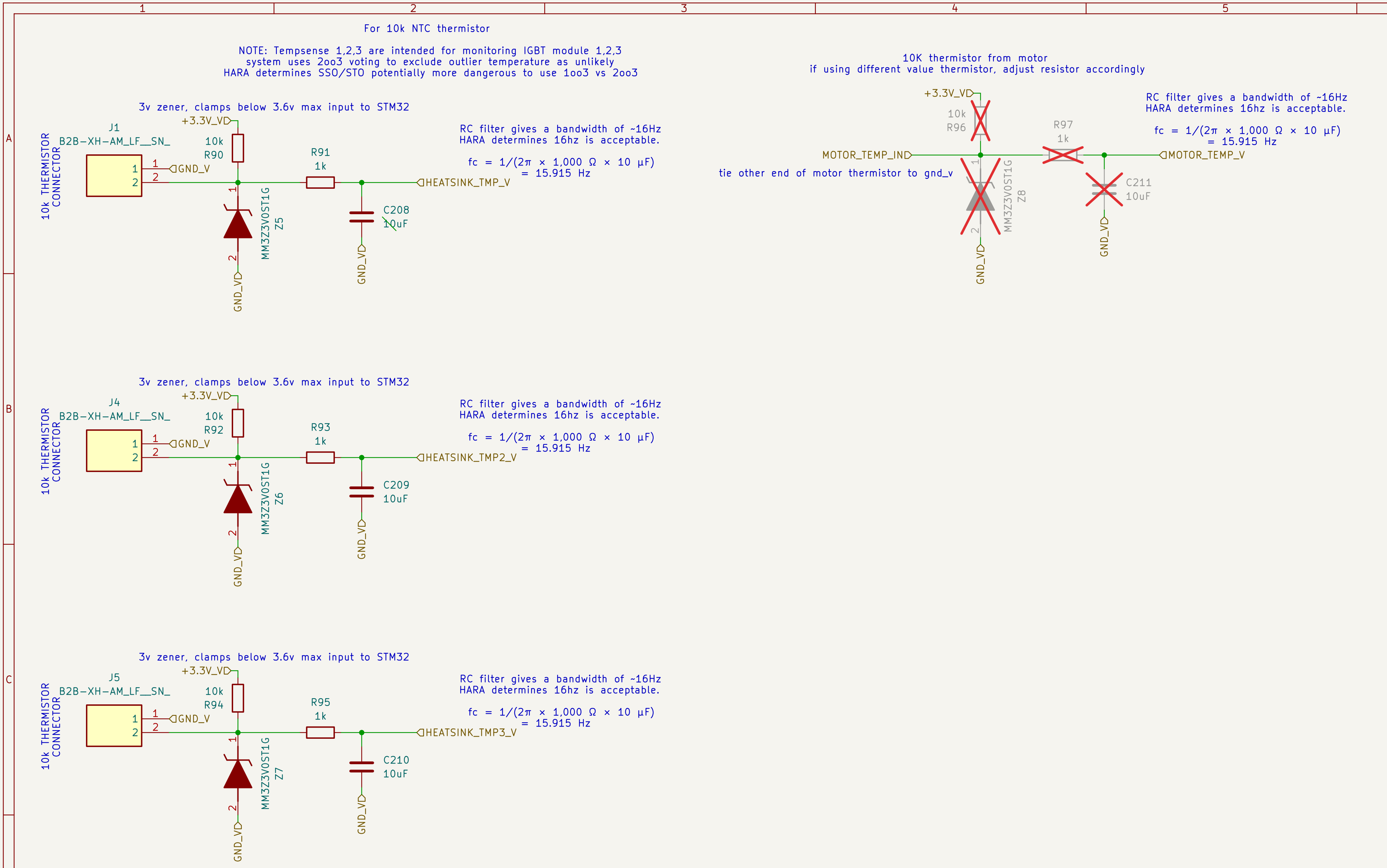
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 20/21

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Power/
File: Power.kicad_sch

Title: InverterGen5 - Control Board

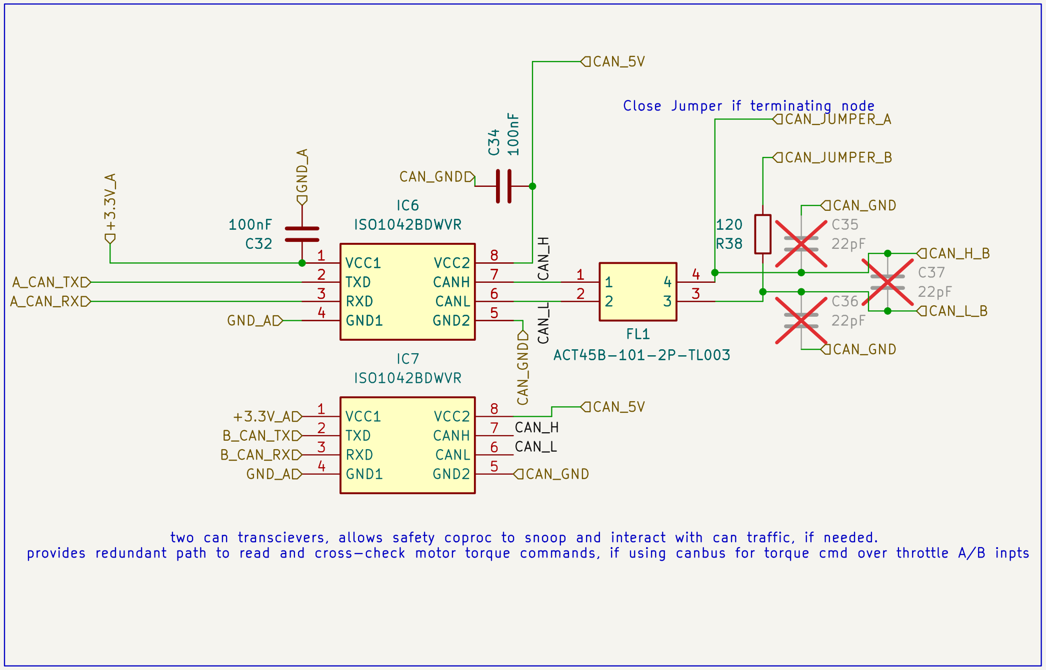
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 9/21

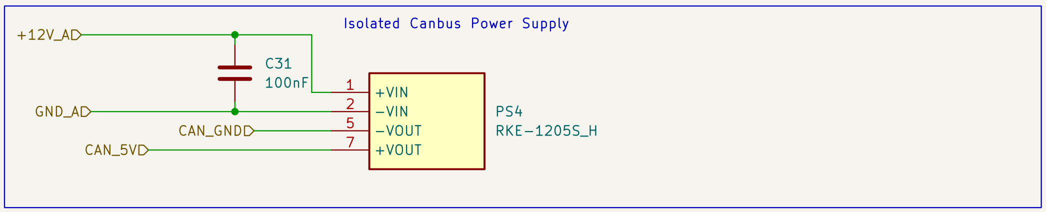


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 9/21

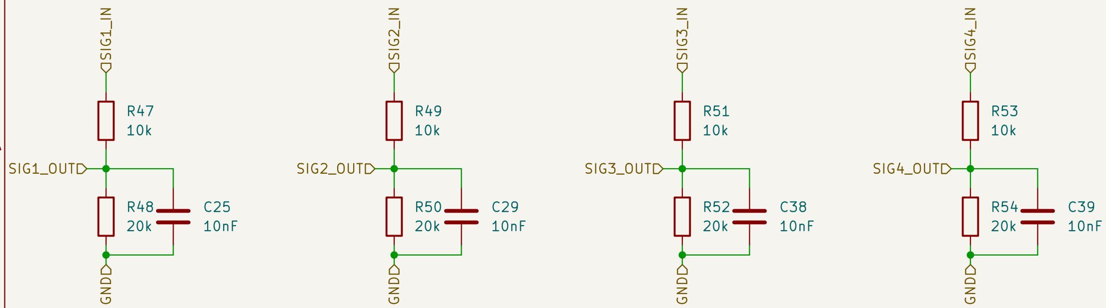




HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 11/21

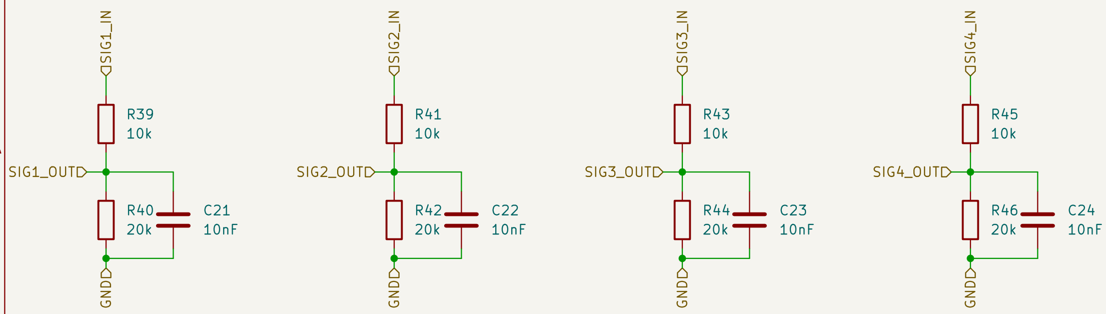


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

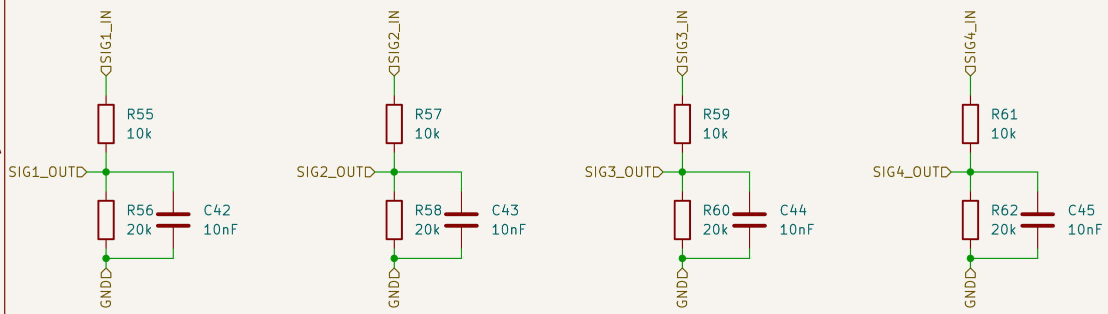
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21



HW–C2–PCB–CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

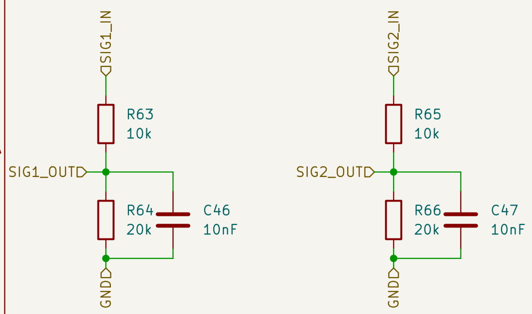
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: InverterGen5 – Control Board

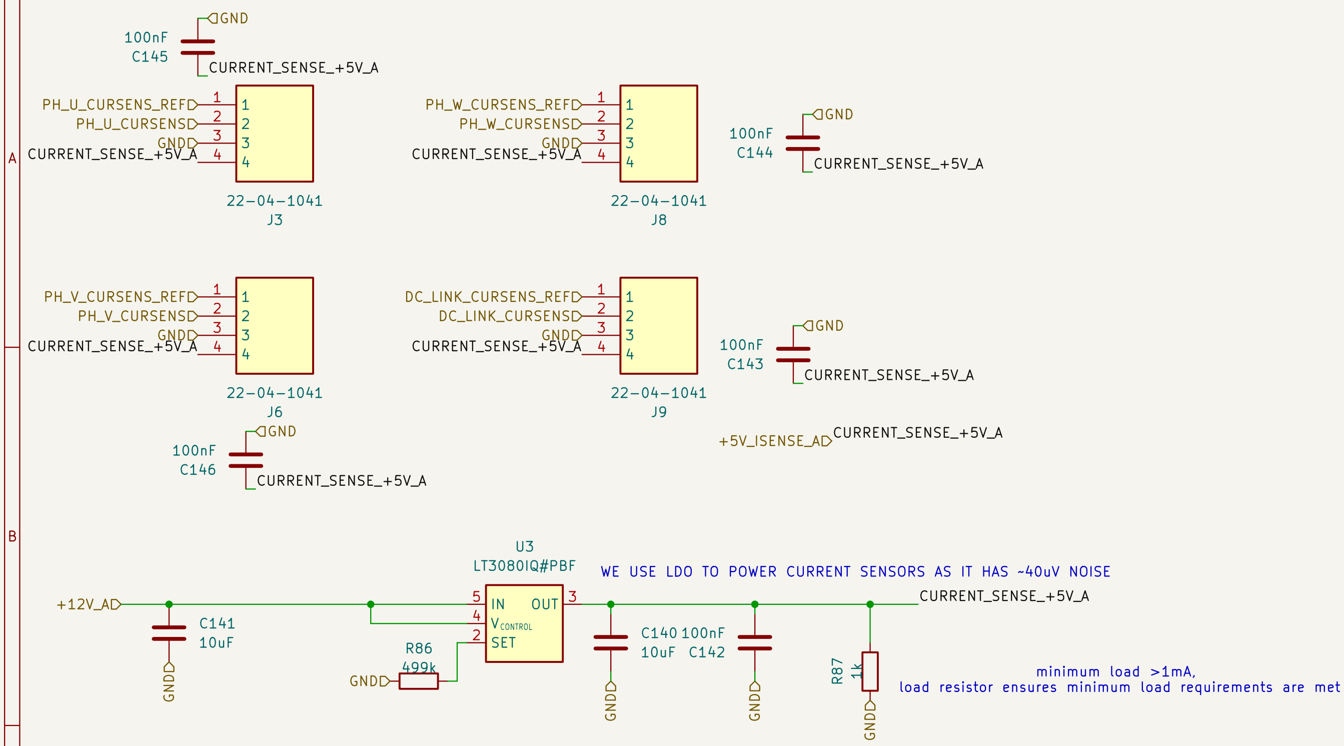
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 14/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch

Title: InverterGen5 – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 15/21

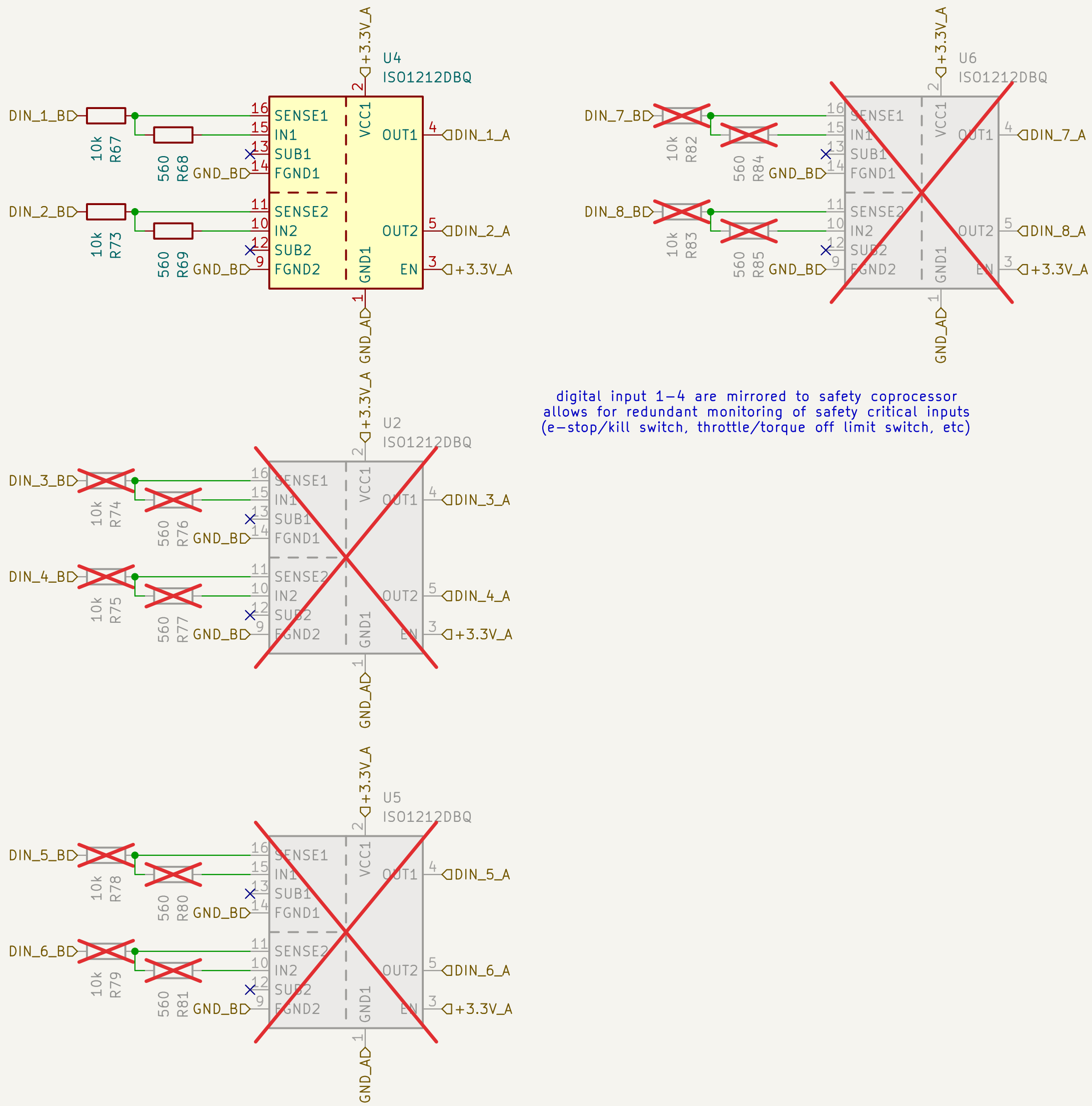


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: InverterGen5 — Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21



digital input 1-4 are mirrored to safety coprocessor
allows for redundant monitoring of safety critical inputs
(e-stop/kill switch, throttle/torque off limit switch, etc)

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

Title: InverterGen5 — Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 18/21